INSTITUTE OF DISTANCE LEARNING

KWAME NKRUMAH UNIVERSITY OF SCIENCE AND TECHNOLOGY, KUMASI, GHANA

ME 392

INDUSTRIAL ENGINEERING AND ERGONOMICS

UNIT 3: FORECASTING

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What is Forecasting?

What is a forecast?



Definition

- Forecast is a statement of what is likely to happen in the future.
- Forecasting may involve taking historical data and projecting them into the future with some mathematical model.
    - Subjective judgements or intuitive prediction.
- Forecasts are seldom perfect
- There is no superior method for all organizations.
- They are also costly and time consuming to prepare.

Types of Forecasts

- ☑ **Economic forecasts**: for predicting e.g. inflation rate, money supply, housing statistic, etc.
- ☑ **Technological forecasts**: for predicting the rate of technological change.
- ☑ **Demand forecasts**: for predicting sales of products.

Strategic Importance of Forecasting

- ☑ The forecast is the only estimate of demand until actual demand becomes known.
- ☑ Forecasts of demand therefore drive decisions in many areas
- ☑ Human Resources – Hiring, training, laying off workers depend on anticipated demand
- ☑ Capacity – Capacity shortages can result in undependable delivery, loss of customers, loss of market share
- ☑ Supply-Chain Management – Good supplier relations and price advantages for materials and parts depend on accurate forecasts
- ☑ Good forecasts are of critical importance in all aspects of a business.

FORECASTING TIME HORIZONS

Short-range forecasts: has time span of up to 1 year but generally less than 3 months. Useful for planning purchasing, job scheduling production levels etc

Intermediate forecasts: generally spans from 3 months up to 3 years. Useful for sales planning, production planning and budgeting.

Long-range forecasts: Generally 3 years or more. Useful for planning new products, facility location or expansion.

Approaches to Forecasting

Two main approaches: **Quantitative & Qualitative**

Quantitative Forecasting: historical data and/causal variables use mathematical models

Qualitative Forecasting: incorporates the factors such as the subjective judgments, intuition, experience and expert knowledge in reaching a forecast.

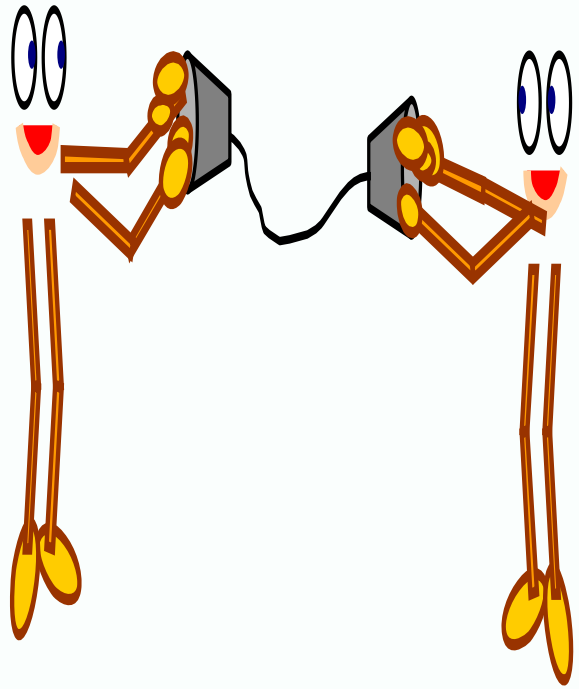
(1) Qualitative Forecasting Methods

These forecasts are also known as judgmental forecasts and they are frequently used to adjust quantitative forecasts.
These methods include:

- **Jury of Executive Opinion:** makes use of the opinions of a small group of high-level managers, often in combination with statistical models
- **Sales Force Composite**: Each sales person estimates what sales will be in his/her region, forecasts reviewed to ensure it is realistic, and then combined at the district and national levels to reach an overall forecast
- **Delphi Method:** Makes use of opinions of expert who may be located at different places (process iterative).
- **Consumer Market Surveys:** Solicits input from customers or potential customers regarding their future purchasing plans.

When Do we use Qualitative Forecasting Approaches

- ☑ Used when situation is vague, there is high level of uncertainty and where little data exist
  - ☑ New products
  - ☑ New technology
  - ☑ Times of Economic and political upheaval



- ☑ Involves intuition, experience
  - ☑ e.g., forecasting sales on Internet

Quantitative Methods Focus

- ☑ Used when situation is 'stable' and historical data exist
  - ✓ Existing products
  - ✓ Current technology
- ☑ Also used when forecasting for the short to medium term when certainty is high
- ☑ Involves mathematical techniques
  - ☑ e.g., forecasting sales of color televisions

Quantitative Approaches

1. Naive approach
2. Moving averages
3. Exponential smoothing
4. Trend projection

Time-Series Models

5. Linear regression

Causal Models

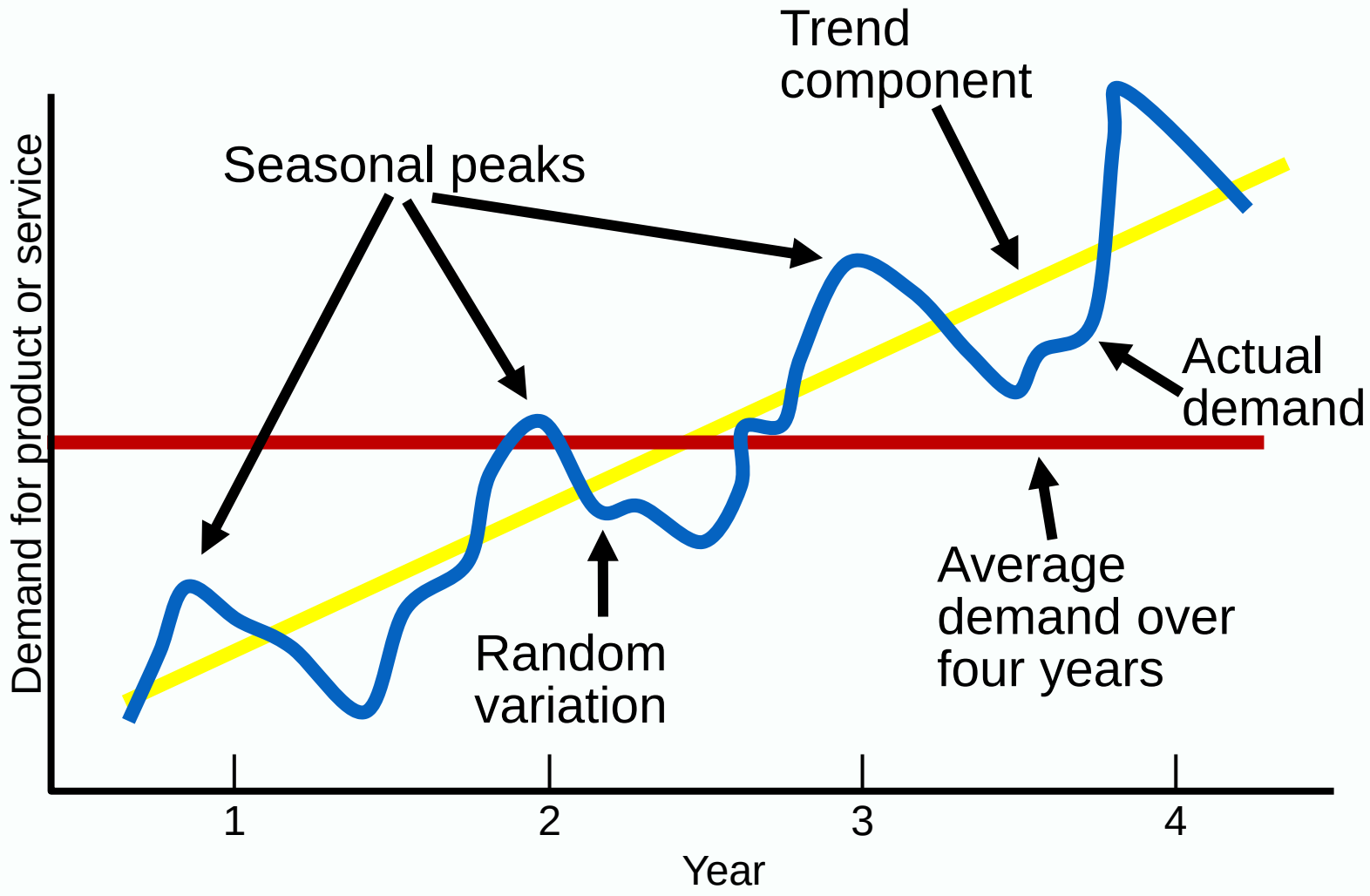
DECOMPOSITION OF TIMES SERIES

- Depends on a series of evenly spaced data points
- Depends on historical data
- Weekly, monthly etc
- Any other variable not showing up in past data is ignored

Decomposition of Times Series:

- **Trend (T)** – gradual upward or downward movement of the data over time.
- **Seasonality (S)-** a data cycle that repeats itself after a period
- **Cycles (C)** – patterns in the data that occur every several years
- **Random variations (R)-** blips in the data caused by chance and unusual situations

Components of Demand



Naive Approach

- ☑ Assumes demand in next period is the same as demand in most recent period
  - ☑ e.g., If May sales were 48, then June's forecast will be 48
- ☑ Sometimes cost effective and efficient
- ☑ Good as a starting forecasting method

Moving Average Method

- assumes that demand in the next period is the average demand over a specified number of periods
  - Useful if we can assume that market demand will be fairly stable over time.
- Moving Average = $\frac{\sum \text{Demand in previous } n \text{ period}}{n}$
- Where **n** is the number of periods

Example of Moving Average

Month	Actual Shed Sales	3-Month Moving Average
January	10	
February	12	
March	13	
April	16	
May	19	
June	23	
July	26	

10
12
13

(10 + 12 + 13)/3 = 11 $^{2}/_{3}$

Weighted Moving Averages

When there is a trend or a pattern, weights can be used to place more emphasis on recent values.

•This makes the technique more responsive to changes.

Weighted Moving Average = $\frac{\sum (\text{weight for period n}) (\text{Demand in previous } n \text{ months})}{\text{Sum of weights}}$

Weighted Moving Average: An Example

Coke sales at Oguaa Retail Store are shown in Table A. Table B shows the weights to be applied to different months. Use this information to conduct:

•A three-month weighted moving average.

Table A

Month	Coke sales
January	10
February	15
March	16
April	12
May	21
June	28
July	30
August	14
September	13
October	16
November	11
December	19

Table B

Weights Applied	Period
3	Last month ago
2	Last two months ago
1	Last three months ago
6	**Total**

Weighted Moving Average: An Example

Month	Demand
1	120
2	110
3	90
4	115
5	125
6	117
7	121
8	126
9	132
10	128

The demand for defense machinery for a certain project is given each month in the table. The defense officer is asked to forecast the demand for the 11th month using three period moving average technique.

The defense officer has decided to use a weighting scheme of **0.5, 0.3, 0.2** and calculated the weighted moving average for the 11th month as follows.

Weighted MA(3):

$$F_{11} = 0.5(128) + 0.3(132) + 0.2(126) = 64 + 39.6 + 25.2 = 128.2$$

Exponential Smoothing

A type of moving average that requires little historical data.

- New forecast = last period's forecast + α (last period's actual demand – last period's forecast)
- $F_t = F_{t-1} + \alpha (A_{t-1} - F_{t-1})$
- α is a weight or smoothing constant that has values between 0 and 1 inclusive
- Requires smoothing constant (α)
  - Ranges from 0 to 1--- $\alpha \leq x \leq 1$
  - Subjectively chosen

Example of Exponential Smoothing

Predicted demand = Balls of Kenkey = 142

Actual demand = 153

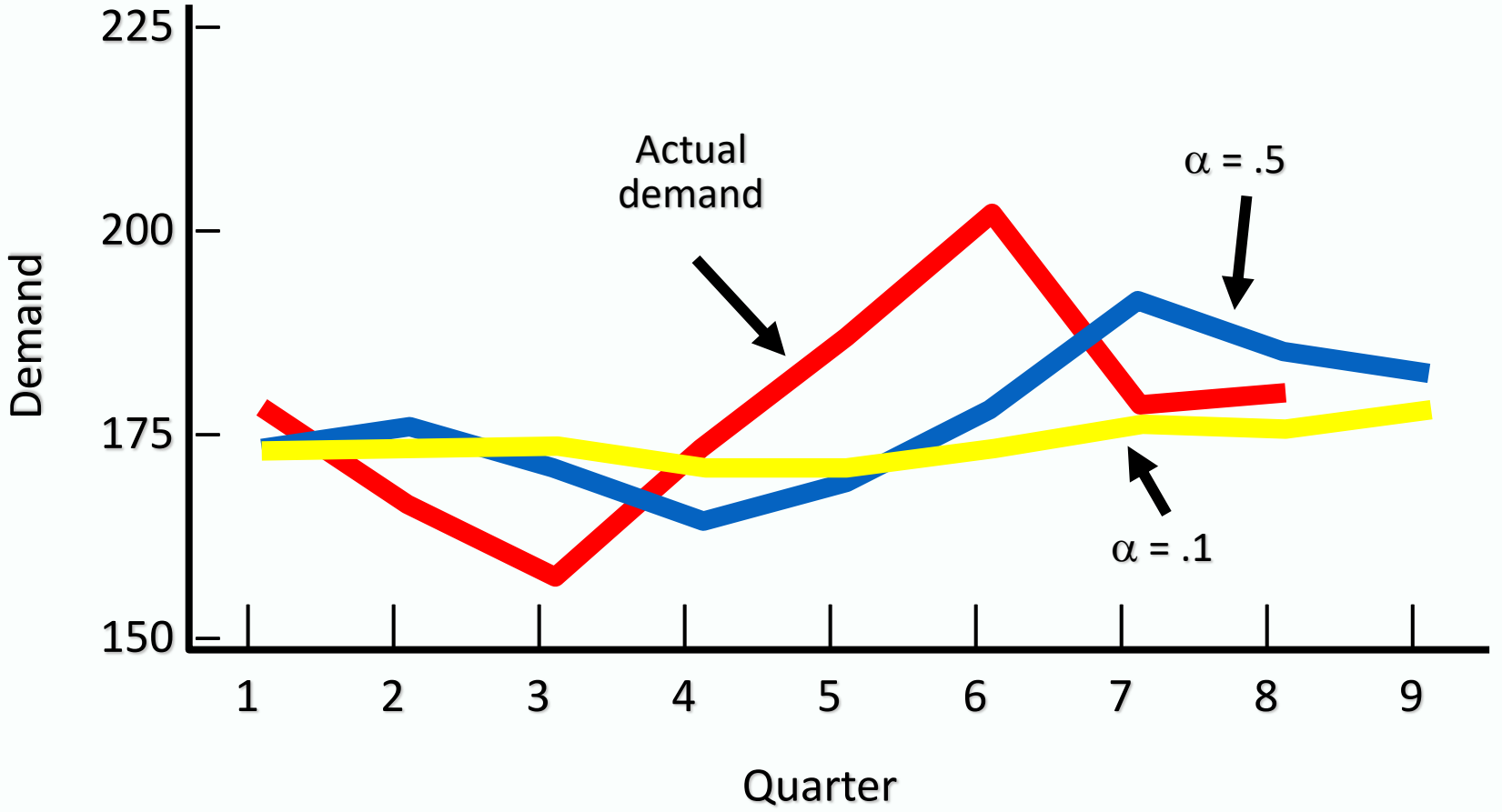
Smoothing constant a = 0.20

What is the forecast??

Effect of Smoothing Constants

Smoothing Constant	Weight Assigned to: Most Recent Period (a)	2nd Most Recent Period $a(1 - a)$	3rd Most Recent Period $a(1 - a)^2$	4th Most Recent Period $a(1 - a)^3$	5th Most Recent Period $a(1 - a)^4$
a = 0.1	0.1	0.09	0.081	0.073	0.066
a =0 .5	0.5	0.25	0.125	0.063	0.031

Impact of Different α



Choosing α

The objective is to obtain the most accurate forecast no matter the technique.

We generally do this by selecting the model that gives us the lowest forecast error

$$\textit{Forecast error} = \textit{Actual demand} - \textit{Forecast value}$$

$$= A_t - F_t$$

Common Measures of Error

(i) Mean Absolute Deviation (MAD)

$$MAD = \frac{\sum |actual - forecast|}{n}$$

(ii) Mean Squared Error (MSE)

$$MSE = \frac{\sum (forecast\ errors)^2}{n}$$

Common Measures of Error…

(iii) Mean Absolute Percent Error (MAPE)

$$MAPE = \frac{100 \sum_{i=1}^{n} |actual_i - forecast_i| / actual_i}{n}$$

Comparison of Forecast Error

Quarter	Actual Tonnage Unloaded	Rounded Forecast with $\alpha = .10$	Absolute Deviation for $\alpha = .10$	Rounded Forecast with $\alpha = .50$	Absolute Deviation for $\alpha = .50$
1	180	175	5	175	5
2	168	176	8	178	10
3	159	175	16	173	14
4	175	173	2	166	9
5	190	173	17	170	20
6	205	175	30	180	25
7	180	178	2	193	13
8	182	178	4	186	4
			84		100

Comparison of Forecast Error

$$MAD = \frac{\sum |\text{deviations}|}{n}$$

For α = 0 .10

= 84/8 = 10.50

For α = .50

= 100/8 = 12.50

e n	Rounded Forecast with α = .50	Absolute Deviation for α = .50
	175	5
	178	10
	173	14
	166	9
	170	20
	180	25
	193	13
8 182 178	186	4
		84
		100

Trend Projections

Fits a line to historical data and then projects it into the future for medium-to-long-range forecasts.

Several mathematical trend equations can be developed (e.g. linear, exponential or quadratic). The most popular ones make use of a linear trend line and *least squares method* .

This approach will yield a straight line that minimises the sum of the squares of the vertical differences from the line to each of the actual observation.

Least square lines are described by $\hat{y} = a + bx$

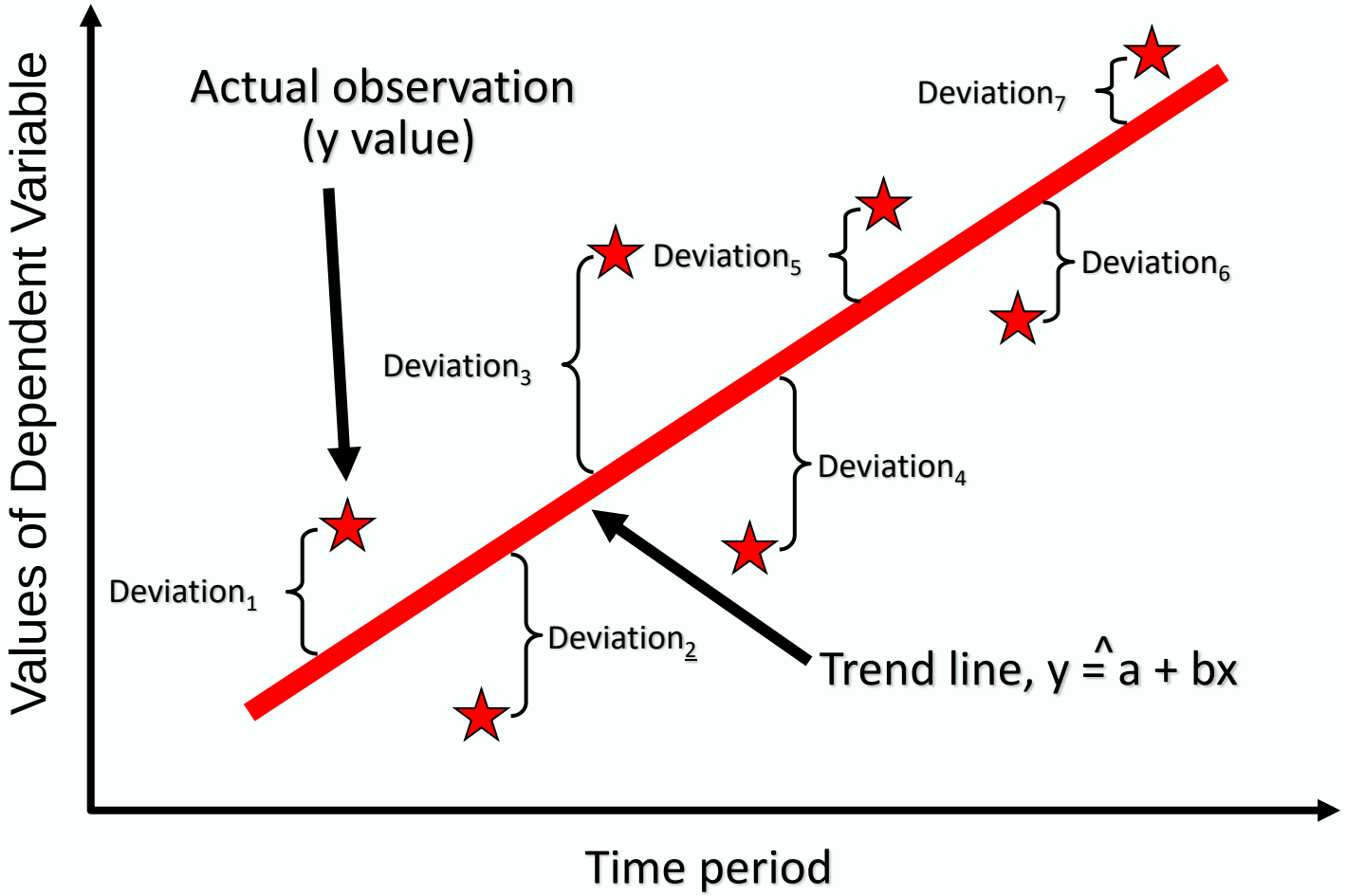
$\hat{Y}$ value to be computed

a is the y-axis intercept

b is the slope of the regression line or rate of change in y

X is the independent variable, which is time in this case

Least Squares Method



Least Squares Method

Equations to calculate the regression variables

$$\hat{y} = a + bx$$

$$b = \frac{\Sigma xy - n\bar{x}\bar{y}}{\Sigma x^2 - n\bar{x}^2} \quad \text{OR} \quad b = \frac{n(\Sigma xy) - (\Sigma x)(\Sigma y)}{n(\Sigma x^{2)} - (\Sigma x)^2}$$

$$a = \bar{y} - b\bar{x} \quad \text{OR} \quad \frac{\Sigma y - b\Sigma x}{n}$$

Least Squares Example

Year	Time Period (x)	Electrical Power Demand	x^2	xy
1999	1	74	1	74
2000	2	79	4	
2001	3	80		240
2002	4	90	16	
2003	5	105		525
2004	6	142	36	
2005	7	122	49	
	$\Sigma x = 28$	$\Sigma y = 692$	$\Sigma x^2 = ???$	$\Sigma xy = ???$
	$\bar{x} = ?$	$\bar{y} = ????$		

$$b = \frac{\Sigma xy - n\bar{x}\bar{y}}{\Sigma x^2 - n\bar{x}^2} = ????$$

$$a = \bar{y} - b\bar{x} = ???$$

Causal Forecasting

Used when changes in one or more independent variables can be used to predict the changes in the dependent variable

Most common technique is linear regression analysis

We apply this technique just as we did in the time series example

Causal Forecasting

Forecasting an outcome based on predictor variables using the least squares technique

$$\hat{y} = a + bx$$

where $\hat{y}$ = computed value of the variable to be predicted (dependent variable)

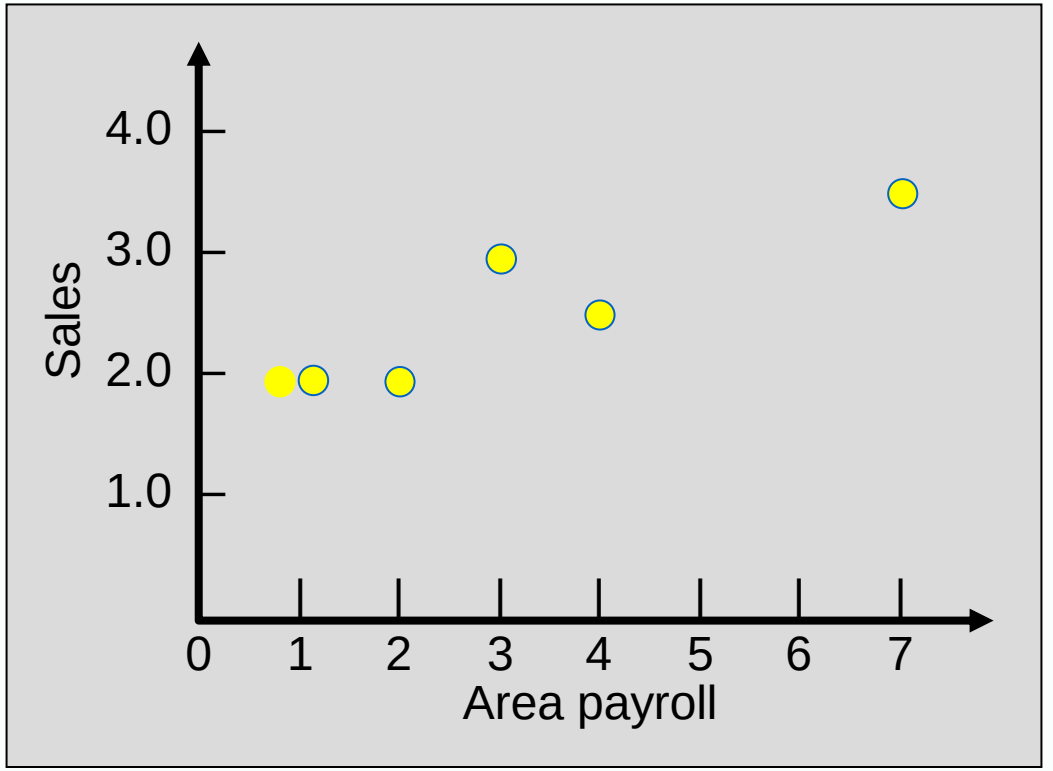
a = y-axis intercept

b = slope of the regression line

x = the independent variable though to predict

Causal Forecasting Example

Sales (GHS000,000), y	Area Payroll (GHS000,000,000), x
2.0	1
3.0	3
2.5	4
2.0	2
2.0	1
3.5	7



Causal Forecasting Example

Sales, y	Payroll, x	x^2	x y
2.0	1	1	2.0
3.0	3	9	9.0
2.5	4	16	10.0
2.0	2	4	4.0
2.0	1	1	2.0
3.5	7	49	24.5
$\sum y = 15.0$	$\sum x = 18$	$\sum x^2 = 80$	$\sum x\,y = 51.5$

$\bar{x} = \sum x/6 = 18/6 = 3$

$\bar{y} = \sum y/6 = 15/6 = 2.5$

$$b = \frac{\sum xy - n\bar{x}\bar{y}}{\sum x^2 - n\bar{x}^2} \qquad \frac{51.5 - (6)(3)(2.5)}{80 - (6)(3^2)} = 0.25$$

$$a = \bar{y} - bx = \quad 2.5 - (0.25)(3) = 1.75$$

Associative Forecasting Example

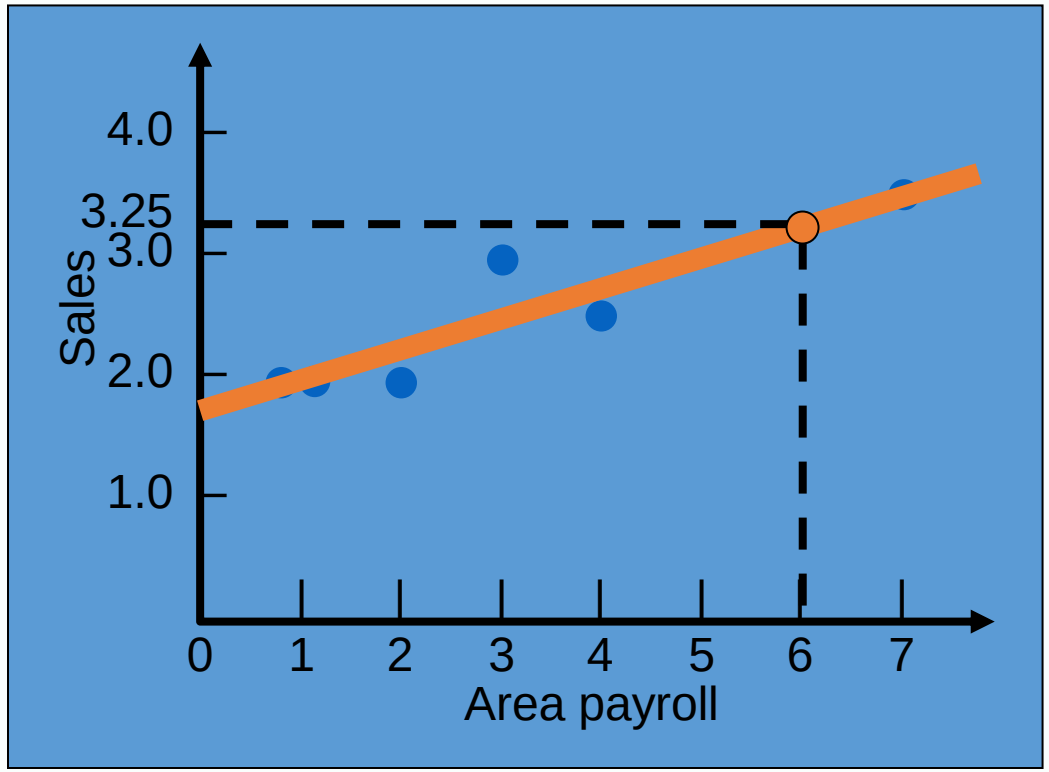
$\hat{y} = 1.75 + .25x$

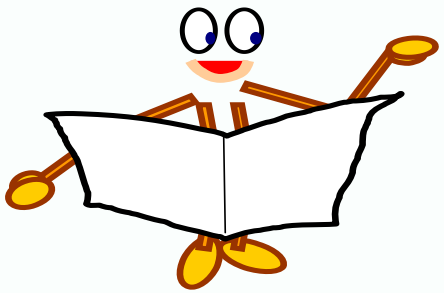
Sales = 1.75 + 0.25(payroll)

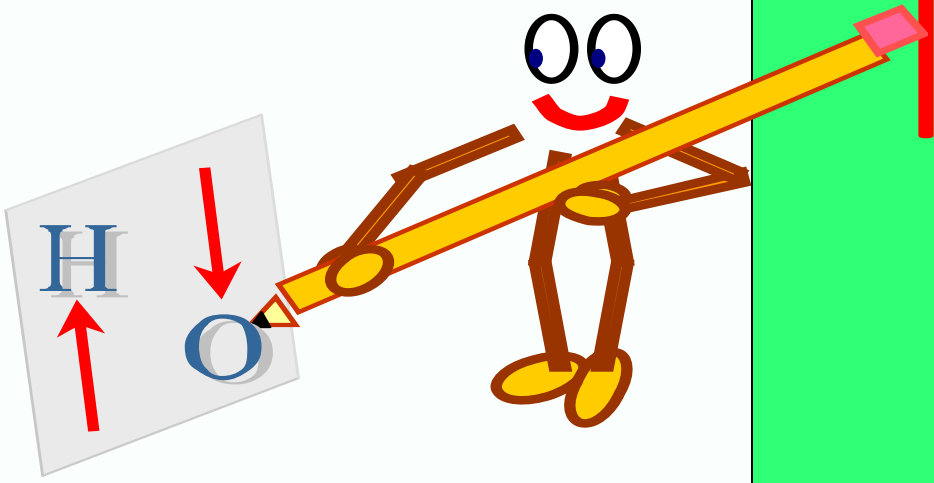
If payroll next year is estimated

to be GHS600 million, then:

Sales = 1.75 + .25(6)
Sales = GHS325,000







Thank you